

Gempack: Its features and capabilities

Introduction to CGE Modeling

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Outline

**Gempack: Its
features and
capabilities**

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Outline

Gempack

Windows
Program

Computation

- Overview of Gempack
- The Gempack Windows programs
- Gempack's way of solving CGE models

Overview

Gempack: Its
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capabilities

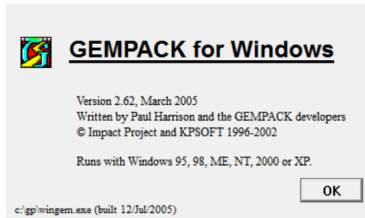
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- Gempack (General Equilibrium Modelling Package) A suite of general-purpose economic modelling software.
- Designed for general equilibrium models (unlike GAMS which is designed for any algebraic mathematical modeling)
- Developed by Centre for Policy Studies (CoPS) Monash University, Australia.

Overview (cont.)

- GEMPACK software is being used in over 400 organizations around the world (universities, government departments and private sector firms), about two thirds of which are outside Australia. It is used to implement and solve a number of economic models.
- The software is user-friendly, and fully documented from a user's point of view.
- GEMPACK provides a number of features designed to make this sort of modelling easier.
- GEMPACK runs on a variety of different machines.

Source: <https://www.copsmodels.com/gempack.htm>

The Gempack Programs provides:

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features and
capabilities

Komarulzaman
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Outline

Gempack

Windows
Program

Computation

- a simple language in which to describe and document the equations of your economic model;
- a program which converts the equations of your model to a form ready for running simulations with the model;
- options for varying the choice of exogenous and endogenous variables and the variables shocked;
- utility programs to assist in managing the database on which the model is based. The data can be inspected, modified, converted to spreadsheets or moved to different machines (including those with different operating systems).

Gempak Windows program

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features and
capabilities

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Outline

Gempack

Windows
Program

Computation

- ❶ WinGEM $\Rightarrow$ Windows interface to GEMPACK
 - ❷ TABmate $\Rightarrow$ Windows text editor for developing TABLO Input files
 - ❸ ViewHAR $\Rightarrow$ Windows program for looking at data in a Header Array file
 - ❹ ViewSOL $\Rightarrow$ Windows program for looking at Solution files
 - ❺ AnalyseGE $\Rightarrow$ Windows program assisting modellers to analyse their results
 - ❻ RunGEM $\Rightarrow$ Windows program for automating simulations with models
 - ❼ RunDynam $\Rightarrow$ Windows interface for recursive dynamic models
 - ❽ Charter $\Rightarrow$ Windows program for drawing graphs
- *We are going to use mostly 1-4! We will use 5 and 10 sometimes.*

Four most often used Gempack Windows program in this course

Gempack: Its
features and
capabilities

Komarulzaman
et al

Outline

Gempack

Windows
Program

Computation



Wingem



Tabmate



Viewhar



Viewsol

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- ④ ViewSQL $\Rightarrow$ Windows program for looking at Solution files

Gempack's way of solving CGE models

- CGE model is simply a system of non-linear equations of n variables and n equations. GEMPACK solves the system.
- How? Simply through Matrix Inversion.
- In GEMPACK we work with *linearized equations*, the CGE model become a system of linear equations.
- But, that's not the only thing GEMPACK do. Gempack correct the linearization error such that the solution is as if solved through non-linear computation.

GEMPACK Correcting linearisation error

Gempack: Its
features and
capabilities

Komarulzaman
et al

Outline

Gempack

Windows
Program

Computation

